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Corolario sobre la regularidad de la convolución

Corolario 1. *Sea* $f \in \mathcal{L}^1(\mathbb{R}^n)$ *y* $g \in \mathcal{C}^k(\mathbb{R}^n)$ *tal que* $\forall \alpha \in (\mathbb{N} \cup \{0\})^n$ *con* $|\alpha| \leqslant k$ *se tiene que* $\|D_x^\alpha g\|_\infty < \infty$. *Entonces,*

$$f * g \in \mathcal{C}^k(\mathbb{R}^n) \quad \wedge \quad \forall \alpha \in (\mathbb{N} \cup \{0\})^n : |\alpha| \leqslant k : D_x^\alpha (f * g) = f * (D_x^\alpha g).$$

***Demostración*:** Basta con probar el caso $\alpha = (1, 0, \ldots, 0)$ y el resto se tiene por inducción.

Sea $x \in \mathbb{R}^n$ y definimos $F(x_1) = (g * f)(x) = \int_{\mathbb{R}^n} g(x-y)\, f(y)\, \mathrm{d}y$ donde $(x_2, \ldots, x_n)$ son fijos. Queremos demostrar que

$$F'(x_1) = \frac{\partial}{\partial x_1} \int_{\mathbb{R}^n} g(x-y)\, f(y)\, \mathrm{d}y = \int_{\mathbb{R}^n} \frac{\partial g}{\partial x_1}(x-y)\, f(y)\, \mathrm{d}y.$$

Aplicamos el `teo-derivacion-bajo-el-signo-integral`/Teorema 1 con $X = \mathbb{R}^n$, μ es la medida de Lebesgue, $t = x_1 \in [a, b]$ arbitrario, y $f(y, x_1) = g(x_1, x_2, \ldots, x_n - y)\, f(y) = g(x-y)\, f(y)$.

Debemos verificar que $\left| \frac{\partial}{\partial x_1} [g(x-y) f(y)] \right| \leqslant g(y) \in \mathcal{L}^1(\mathbb{R}^n)$ donde g es la mayorante:

$$\left| \frac{\partial g(x-y) f(y)}{\partial x_1} \right| = \left| \frac{\partial g}{\partial x_1}(x-y)\, f(y) \right| \leqslant \left\| \frac{\partial g}{\partial x_1} \right\|_\infty \cdot |f(y)| \leqslant M \cdot |f(y)| \in \mathcal{L}^1(\mathbb{R}^n),$$

donde $M = \left\| \frac{\partial g}{\partial x_1} \right\|_\infty < \infty$ por hipótesis. Por tanto, obtenemos

$$\frac{\partial}{\partial x_1} \int_{\mathbb{R}^n} g(x-y)\, f(y)\, \mathrm{d}y = \int_{\mathbb{R}^n} \frac{\partial g}{\partial x_1}(x-y)\, f(y)\, \mathrm{d}y = \frac{\partial g}{\partial x_1} * f(x) = f * \frac{\partial g}{\partial x_1}(x),$$

donde la última igualdad se sigue de la conmutatividad de la convolución. ■

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